Aurora: Module Relay x8

Stakeholders Questions

Prepared for

Technical Staff

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Disclosure is prohibited

Revision History

Version	Date	Author	Description
0.1.0	2026-07-02	Martin Pellizzer	Starting version

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Questions

Question 1

Field	Value
Stakeholder ID	1
Stakeholder Role	System Architect
Question ID	1
Question Text	What exactly is the responsibility of the Relay Module within Aurora?

Question 2

Field	Value
Stakeholder ID	1
Stakeholder Role	System Architect
Question ID	2
Question Text	Which functions of the overall Aurora system are allocated specifically to the Relay Module?

Question 3

Field	Value
Stakeholder ID	1
Stakeholder Role	System Architect
Question ID	3
Question Text	What behavior does the system require from each of the eight relay outputs?

Question 4

Field	Value
Stakeholder ID	1
Stakeholder Role	System Architect
Question ID	4
Question Text	What must happen to every relay output when communication with the CORE is lost?

Question 5

Field	Value
Stakeholder ID	1
Stakeholder Role	System Architect
Question ID	5
Question Text	What is the required system behavior if the Relay Module itself fails?

Question 6

Field	Value
Stakeholder ID	2
Stakeholder Role	CORE Firmware Engineer
Question ID	6
Question Text	Which responsibilities belong to the CORE and which responsibilities belong to the Relay Module?

Question 7

Field	Value
Stakeholder ID	2
Stakeholder Role	CORE Firmware Engineer
Question ID	7

Field	Value
Question Text	What commands must the CORE be able to send to the Relay Module?

Question 8

Field	Value
Stakeholder ID	2
Stakeholder Role	CORE Firmware Engineer
Question ID	8
Question Text	What information must the Relay Module provide back to the CORE?

Question 9

Field	Value
Stakeholder ID	2
Stakeholder Role	CORE Firmware Engineer
Question ID	9
Question Text	How should the CORE detect that communication with the Relay Module has been lost?

Question 10

Field	Value
Stakeholder ID	2
Stakeholder Role	CORE Firmware Engineer
Question ID	10
Question Text	What should the CORE do when communication with the Relay Module is lost?

Question 11

Field	Value
Stakeholder ID	3
Stakeholder Role	Electrical Engineer
Question ID	11
Question Text	What are the electrical specifications of the eight relay-driver outputs, based on the physical relays they must control?

Question 12

Field	Value
Stakeholder ID	3
Stakeholder Role	Electrical Engineer
Question ID	12
Question Text	What are the electrical characteristics of the physical relay coils connected to each of the eight driver outputs?

Question 13

Field	Value
Stakeholder ID	3
Stakeholder Role	Electrical Engineer
Question ID	13
Question Text	What is the required input supply voltage range, including tolerances and transient conditions?

Question 14

Field	Value
Stakeholder ID	3
Stakeholder Role	Electrical Engineer
Question ID	14

Field	Value
Question Text	What protection is required to handle the inductive voltage transient generated when a relay coil is de-energized?

Question 15

Field	Value
Stakeholder ID	3
Stakeholder Role	Electrical Engineer
Question ID	15
Question Text	What electrical behavior is required during power loss, brownout, and restoration of power?

Question 16

Field	Value
Stakeholder ID	4
Stakeholder Role	Safety Engineer
Question ID	16
Question Text	What safety functions, if any, are assigned specifically to the Relay Module?

Question 17

Field	Value
Stakeholder ID	4
Stakeholder Role	Safety Engineer
Question ID	17
Question Text	What is the required safe state for each of the eight relay outputs?

Question 18

Field	Value
Stakeholder ID	4
Stakeholder Role	Safety Engineer
Question ID	18
Question Text	What shall happen to each relay output when the Relay Module detects an internal fault?

Question 19

Field	Value
Stakeholder ID	4
Stakeholder Role	Safety Engineer
Question ID	19
Question Text	What shall happen to each relay output when communication with the CORE is lost?

Question 20

Field	Value
Stakeholder ID	4
Stakeholder Role	Safety Engineer
Question ID	20
Question Text	Which safety decisions must be made by the CORE, and which must be enforced locally by the Relay Module?

Question 21

Field	Value
Stakeholder ID	5
Stakeholder Role	Process Engineer
Question ID	21

Field	Value
Question Text	What process functions are controlled or influenced by the eight relay outputs of the Relay Module?

Question 22

Field	Value
Stakeholder ID	5
Stakeholder Role	Process Engineer
Question ID	22
Question Text	What is the required sequence of operations for starting ozone generation?

Question 23

Field	Value
Stakeholder ID	5
Stakeholder Role	Process Engineer
Question ID	23
Question Text	What is the required sequence of operations for stopping ozone generation?

Question 24

Field	Value
Stakeholder ID	5
Stakeholder Role	Process Engineer
Question ID	24
Question Text	What conditions must be satisfied before ozone generation is permitted to start?

Question 25

Field	Value
Stakeholder ID	5
Stakeholder Role	Process Engineer
Question ID	25
Question Text	What process conditions require ozone generation to stop immediately?